Simple LSTM Forward Pass with Full Intermediate Steps
(5 Time Steps, Two Stacked LSTM Units — Anomaly Detection Network)

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This version extends the single-unit example to a small **anomaly detection network** built from **two stacked LSTM units**. Unit 1 reads the raw input signal x_t exactly as before — its section below is unchanged from the original document. Unit 2 sits on top of Unit 1: at every time step its input is Unit 1's hidden state $h_t^{(1)}$. The final output of the stack, $h_t^{(2)}$, is the network's prediction of the "normal" signal value; at detection time, a large squared error between $h_5^{(2)}$ and the expected normal value flags the sequence as anomalous. Every single arithmetic operation is shown explicitly for each gate and state update in both units.

Part I: First LSTM Unit (Unchanged from the Original Document)

Weights (Randomly Initialized, Shared Across All Time Steps)

Using the naming convention from our previous work:

Forget gate

- $w_{hf} = 0.28$
- $w_{xf} = -0.15$
- $b_f = 0.05$

Input gate (sigmoid)

- $w_{hi1} = 0.35$
- $w_{xi1} = 0.42$
- $b_{i1} = 0.18$

Candidate

- $w_{hi2} = 0.22$
- $w_{xi2} = 0.31$
- $b_{i2} = 0.09$

Output gate

- $w_{ho} = 0.47$
- $w_{xo} = 0.19$
- $b_o = 0.26$

Input Sequence

- $x_1 = 0.5$
- $x_2 = 0.7$

- $x_3 = 0.4$
- $x_4 = 0.8$
- $x_5 = 0.6$

Initial State

- $h_0 = 0$
- $C_0 = 0$

The Six Equations Used at Every Time Step

$$
\begin{aligned}
f_t &= \sigma(w_{hf} \cdot h_{t-1} + w_{xf} \cdot x_t + b_f) \\
i_t &= \sigma(w_{hi1} \cdot h_{t-1} + w_{xi1} \cdot x_t + b_{i1}) \\
\tilde{C}_t &= \tanh(w_{hi2} \cdot h_{t-1} + w_{xi2} \cdot x_t + b_{i2}) \\
o_t &= \sigma(w_{ho} \cdot h_{t-1} + w_{xo} \cdot x_t + b_o) \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\
h_t &= o_t \odot \tanh(C_t)
\end{aligned}
$$

(where $\sigma(z) = \frac{1}{1+e^{-z}}$ and tanh is the standard hyperbolic tangent function).

Full Forward Pass with Every Intermediate Step

Time Step $t = 1$

Inputs: $h_0 = 0$, $C_0 = 0$, $x_1 = 0.5$

Forget gate

- Linear combination: $0.28 \times 0 + (-0.15) \times 0.5 + 0.05 = -0.075 + 0.05 = -0.025$
- $f_1 = \sigma(-0.025) = \frac{1}{1+e^{0.025}} \approx \frac{1}{2.025315} \approx 0.4938$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0 + 0.42 \times 0.5 + 0.18 = 0.21 + 0.18 = 0.39$
- $i_1 = \sigma(0.39) \approx 0.5963$

Candidate

- Linear combination: $0.22 \times 0 + 0.31 \times 0.5 + 0.09 = 0.155 + 0.09 = 0.245$
- $\tilde{C}_1 = \tanh(0.245) \approx 0.2404$

Output gate

- Linear combination: $0.47 \times 0 + 0.19 \times 0.5 + 0.26 = 0.095 + 0.26 = 0.355$
- $o_1 = \sigma(0.355) \approx 0.5878$

Cell state update

- $C_1 = (0.4938 \times 0) + (0.5963 \times 0.2404) = 0 + 0.1434 = 0.1434$

Hidden state update

- $h_1 = 0.5878 \times \tanh(0.1434) \approx 0.5878 \times 0.1426 = 0.0838$

Updated after $t = 1$: $\quad h_1 \approx 0.0838, \quad C_1 \approx 0.1434$

Time Step $t = 2$

Inputs: $h_1 \approx 0.0838$, $C_1 \approx 0.1434$, $x_2 = 0.7$

Forget gate

- Linear combination: $0.28 \times 0.0838 + (-0.15) \times 0.7 + 0.05 = -0.0315$
- $f_2 = \sigma(-0.0315) \approx 0.4921$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.0838 + 0.42 \times 0.7 + 0.18 = 0.5033$
- $i_2 = \sigma(0.5033) \approx 0.6232$

Candidate

- Linear combination: $0.22 \times 0.0838 + 0.31 \times 0.7 + 0.09 = 0.3254$
- $\tilde{C}_2 = \tanh(0.3254) \approx 0.3140$

Output gate

- Linear combination: $0.47 \times 0.0838 + 0.19 \times 0.7 + 0.26 = 0.4324$
- $o_2 = \sigma(0.4324) \approx 0.6064$

Cell state update

- $C_2 = (0.4921 \times 0.1434) + (0.6232 \times 0.3140) = 0.2663$

Hidden state update

- $h_2 = 0.6064 \times \tanh(0.2663) \approx 0.1578$

Updated after $t = 2$: $\quad h_2 \approx 0.1578, \quad C_2 \approx 0.2663$

Time Step $t = 3$

Inputs: $h_2 \approx 0.1578$, $C_2 \approx 0.2663$, $x_3 = 0.4$

Forget gate

- Linear combination: $0.28 \times 0.1578 + (-0.15) \times 0.4 + 0.05 = 0.0342$
- $f_3 = \sigma(0.0342) \approx 0.5086$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.1578 + 0.42 \times 0.4 + 0.18 = 0.4032$
- $i_3 = \sigma(0.4032) \approx 0.5995$

Candidate

- Linear combination: $0.22 \times 0.1578 + 0.31 \times 0.4 + 0.09 = 0.2487$
- $\tilde{C}_3 = \tanh(0.2487) \approx 0.2440$

Output gate

- Linear combination: $0.47 \times 0.1578 + 0.19 \times 0.4 + 0.26 = 0.4102$
- $o_3 = \sigma(0.4102) \approx 0.6011$

Cell state update

- $C_3 = (0.5086 \times 0.2663) + (0.5995 \times 0.2440) = 0.2817$

Hidden state update

- $h_3 = 0.6011 \times \tanh(0.2817) \approx 0.1649$

Updated after $t = 3$: $\quad h_3 \approx 0.1649, \quad C_3 \approx 0.2817$

Time Step $t = 4$

Inputs: $h_3 \approx 0.1649$, $C_3 \approx 0.2817$, $x_4 = 0.8$

Forget gate

- Linear combination: $0.28 \times 0.1649 + (-0.15) \times 0.8 + 0.05 = -0.0238$
- $f_4 = \sigma(-0.0238) \approx 0.4940$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.1649 + 0.42 \times 0.8 + 0.18 = 0.5737$
- $i_4 = \sigma(0.5737) \approx 0.6396$

Candidate

- Linear combination: $0.22 \times 0.1649 + 0.31 \times 0.8 + 0.09 = 0.3743$
- $\tilde{C}_4 = \tanh(0.3743) \approx 0.3580$

Output gate

- Linear combination: $0.47 \times 0.1649 + 0.19 \times 0.8 + 0.26 = 0.4895$
- $o_4 = \sigma(0.4895) \approx 0.6200$

Cell state update

- $C_4 = (0.4940 \times 0.2817) + (0.6396 \times 0.3580) = 0.3681$

Hidden state update

- $h_4 = 0.6200 \times \tanh(0.3681) \approx 0.2184$

Updated after $t = 4$: $\quad h_4 \approx 0.2184, \quad C_4 \approx 0.3681$

Time Step $t = 5$

Inputs: $h_4 \approx 0.2184$, $C_4 \approx 0.3681$, $x_5 = 0.6$

Forget gate

- Linear combination: $0.28 \times 0.2184 + (-0.15) \times 0.6 + 0.05 = 0.0212$
- $f_5 = \sigma(0.0212) \approx 0.5053$

Input gate (sigmoid)

- Linear combination: $0.35 \times 0.2184 + 0.42 \times 0.6 + 0.18 = 0.5084$
- $i_5 = \sigma(0.5084) \approx 0.6244$

Candidate

- Linear combination: $0.22 \times 0.2184 + 0.31 \times 0.6 + 0.09 = 0.3240$
- $\tilde{C}_5 = \tanh(0.3240) \approx 0.3130$

Output gate

- Linear combination: $0.47 \times 0.2184 + 0.19 \times 0.6 + 0.26 = 0.4766$
- $o_5 = \sigma(0.4766) \approx 0.6169$

Cell state update

- $C_5 = (0.5053 \times 0.3681) + (0.6244 \times 0.3130) = 0.3814$

Hidden state update

- $h_5 = 0.6169 \times \tanh(0.3814) \approx 0.2246$

Final updated values after $t = 5$: $\quad h_5 \approx 0.2246, \quad C_5 \approx 0.3814$

This completes the full forward pass of Unit 1. Every intermediate linear combination, activation, and state update has been shown with explicit arithmetic for all 5 time steps using the exact equations we derived. The states h_t and C_t are updated after each step and carried forward to the next time step.

Part II: Second LSTM Unit (Stacked on Top of Unit 1)

Unit 2 uses the same six equations as Unit 1, with two changes:

- Its **input at each time step is Unit 1's hidden state** $h_t^{(1)}$ (computed in Part I), taking the place of x_t.
- It has its **own hidden state** $h_t^{(2)}$, **own cell state** $C_t^{(2)}$, and **own 12 weights**, initialized below.

Superscript (1) refers to Unit 1 and superscript (2) to Unit 2. Unit 1's values in Part I are completely unchanged; Unit 1 never sees Unit 2.

Unit 2 Weights (Randomly Initialized, Shared Across All Time Steps)

Using the same naming convention, with superscript (2):

Forget gate

- $w_{hf}^{(2)} = 0.33$
- $w_{xf}^{(2)} = -0.21$
- $b_f^{(2)} = 0.07$

Input gate (sigmoid)

- $w_{hi1}^{(2)} = 0.29$
- $w_{xi1}^{(2)} = 0.38$
- $b_{i1}^{(2)} = 0.12$

Candidate

- $w_{hi2}^{(2)} = 0.18$
- $w_{xi2}^{(2)} = 0.27$
- $b_{i2}^{(2)} = 0.14$

Output gate

- $w_{ho}^{(2)} = 0.41$
- $w_{xo}^{(2)} = 0.24$
- $b_o^{(2)} = 0.21$

Input Sequence to Unit 2 (= Unit 1 Hidden States from Part I)

- $h_1^{(1)} \approx 0.0838$
- $h_2^{(1)} \approx 0.1578$
- $h_3^{(1)} \approx 0.1649$

- $h_4^{(1)} \approx 0.2184$
- $h_5^{(1)} \approx 0.2246$

Initial State

- $h_0^{(2)} = 0$
- $C_0^{(2)} = 0$

The Six Equations Used at Every Time Step (Unit 2)

$$
\begin{aligned}
f_t^{(2)} &= \sigma(w_{hf}^{(2)} \cdot h_{t-1}^{(2)} + w_{xf}^{(2)} \cdot h_t^{(1)} + b_f^{(2)}) \\
i_t^{(2)} &= \sigma(w_{hi1}^{(2)} \cdot h_{t-1}^{(2)} + w_{xi1}^{(2)} \cdot h_t^{(1)} + b_{i1}^{(2)}) \\
\tilde{C}_t^{(2)} &= \tanh(w_{hi2}^{(2)} \cdot h_{t-1}^{(2)} + w_{xi2}^{(2)} \cdot h_t^{(1)} + b_{i2}^{(2)}) \\
o_t^{(2)} &= \sigma(w_{ho}^{(2)} \cdot h_{t-1}^{(2)} + w_{xo}^{(2)} \cdot h_t^{(1)} + b_o^{(2)}) \\
C_t^{(2)} &= f_t^{(2)} \odot C_{t-1}^{(2)} + i_t^{(2)} \odot \tilde{C}_t^{(2)} \\
h_t^{(2)} &= o_t^{(2)} \odot \tanh(C_t^{(2)})
\end{aligned}
$$

The only structural difference from Unit 1 is that x_t has been replaced by $h_t^{(1)}$.

Full Forward Pass of Unit 2 with Every Intermediate Step

Time Step $t = 1$

Inputs: $h_0^{(2)} \approx 0.0000, \quad C_0^{(2)} \approx 0.0000, \quad$ input from Unit 1: $h_1^{(1)} \approx 0.0838$

Forget gate

- Linear combination: $0.33 \times 0.0000 + (-0.21) \times 0.0838 + 0.07 = 0.0000 + (-0.0176) + 0.07 = 0.0524$
- $f_1^{(2)} = \sigma(0.0524) \approx 0.5131$

Input gate (sigmoid)

- Linear combination: $0.29 \times 0.0000 + 0.38 \times 0.0838 + 0.12 = 0.0000 + 0.0318 + 0.12 = 0.1518$
- $i_1^{(2)} = \sigma(0.1518) \approx 0.5379$

Candidate

- Linear combination: $0.18 \times 0.0000 + 0.27 \times 0.0838 + 0.14 = 0.0000 + 0.0226 + 0.14 = 0.1626$
- $\tilde{C}_1^{(2)} = \tanh(0.1626) \approx 0.1612$

Output gate

- Linear combination: $0.41 \times 0.0000 + 0.24 \times 0.0838 + 0.21 = 0.0000 + 0.0201 + 0.21 = 0.2301$
- $o_1^{(2)} = \sigma(0.2301) \approx 0.5573$

Cell state update

- $C_1^{(2)} = (0.5131 \times 0.0000) + (0.5379 \times 0.1612) = 0.0000 + 0.0867 = 0.0867$

Hidden state update

- $h_1^{(2)} = 0.5573 \times \tanh(0.0867) \approx 0.5573 \times 0.0865 = 0.0482$

Updated after $t = 1$: $\quad h_1^{(2)} \approx 0.0482, \quad C_1^{(2)} \approx 0.0867$

Time Step $t = 2$

Inputs: $h_1^{(2)} \approx 0.0482, \quad C_1^{(2)} \approx 0.0867, \quad$ input from Unit 1: $h_2^{(1)} \approx 0.1578$

Forget gate

- Linear combination: $0.33 \times 0.0482 + (-0.21) \times 0.1578 + 0.07 = 0.0159 + (-0.0331) + 0.07 = 0.0528$
- $f_2^{(2)} = \sigma(0.0528) \approx 0.5132$

Input gate (sigmoid)

- Linear combination: $0.29 \times 0.0482 + 0.38 \times 0.1578 + 0.12 = 0.0140 + 0.0600 + 0.12 = 0.1939$
- $i_2^{(2)} = \sigma(0.1939) \approx 0.5483$

Candidate

- Linear combination: $0.18 \times 0.0482 + 0.27 \times 0.1578 + 0.14 = 0.0087 + 0.0426 + 0.14 = 0.1913$
- $\tilde{C}_2^{(2)} = \tanh(0.1913) \approx 0.1890$

Output gate

- Linear combination: $0.41 \times 0.0482 + 0.24 \times 0.1578 + 0.21 = 0.0198 + 0.0379 + 0.21 = 0.2676$
- $o_2^{(2)} = \sigma(0.2676) \approx 0.5665$

Cell state update

- $C_2^{(2)} = (0.5132 \times 0.0867) + (0.5483 \times 0.1890) = 0.0445 + 0.1036 = 0.1481$

Hidden state update

- $h_2^{(2)} = 0.5665 \times \tanh(0.1481) \approx 0.5665 \times 0.1471 = 0.0833$

Updated after $t = 2$: $\quad h_2^{(2)} \approx 0.0833, \quad C_2^{(2)} \approx 0.1481$

Time Step $t = 3$

Inputs: $h_2^{(2)} \approx 0.0833, \quad C_2^{(2)} \approx 0.1481, \quad$ input from Unit 1: $h_3^{(1)} \approx 0.1649$

Forget gate

- Linear combination: $0.33 \times 0.0833 + (-0.21) \times 0.1649 + 0.07 = 0.0275 + (-0.0346) + 0.07 = 0.0629$
- $f_3^{(2)} = \sigma(0.0629) \approx 0.5157$

Input gate (sigmoid)

- Linear combination: $0.29 \times 0.0833 + 0.38 \times 0.1649 + 0.12 = 0.0242 + 0.0627 + 0.12 = 0.2068$
- $i_3^{(2)} = \sigma(0.2068) \approx 0.5515$

Candidate

- Linear combination: $0.18 \times 0.0833 + 0.27 \times 0.1649 + 0.14 = 0.0150 + 0.0445 + 0.14 = 0.1995$
- $\tilde{C}_3^{(2)} = \tanh(0.1995) \approx 0.1969$

Output gate

- Linear combination: $0.41 \times 0.0833 + 0.24 \times 0.1649 + 0.21 = 0.0342 + 0.0396 + 0.21 = 0.2837$
- $o_3^{(2)} = \sigma(0.2837) \approx 0.5705$

Cell state update

- $C_3^{(2)} = (0.5157 \times 0.1481) + (0.5515 \times 0.1969) = 0.0764 + 0.1086 = 0.1850$

Hidden state update

- $h_3^{(2)} = 0.5705 \times \tanh(0.1850) \approx 0.5705 \times 0.1829 = 0.1043$

Updated after $t = 3$: $\quad h_3^{(2)} \approx 0.1043, \quad C_3^{(2)} \approx 0.1850$

Time Step $t = 4$

Inputs: $h_3^{(2)} \approx 0.1043, \quad C_3^{(2)} \approx 0.1850, \quad$ input from Unit 1: $h_4^{(1)} \approx 0.2184$

Forget gate

- Linear combination: $0.33 \times 0.1043 + (-0.21) \times 0.2184 + 0.07 = 0.0344 + (-0.0459) + 0.07 = 0.0586$
- $f_4^{(2)} = \sigma(0.0586) \approx 0.5146$

Input gate (sigmoid)

- Linear combination: $0.29 \times 0.1043 + 0.38 \times 0.2184 + 0.12 = 0.0303 + 0.0830 + 0.12 = 0.2333$
- $i_4^{(2)} = \sigma(0.2333) \approx 0.5580$

Candidate

- Linear combination: $0.18 \times 0.1043 + 0.27 \times 0.2184 + 0.14 = 0.0188 + 0.0590 + 0.14 = 0.2177$
- $\tilde{C}_4^{(2)} = \tanh(0.2177) \approx 0.2144$

Output gate

- Linear combination: $0.41 \times 0.1043 + 0.24 \times 0.2184 + 0.21 = 0.0428 + 0.0524 + 0.21 = 0.3052$
- $o_4^{(2)} = \sigma(0.3052) \approx 0.5757$

Cell state update

- $C_4^{(2)} = (0.5146 \times 0.1850) + (0.5580 \times 0.2144) = 0.0952 + 0.1196 = 0.2148$

Hidden state update

- $h_4^{(2)} = 0.5757 \times \tanh(0.2148) \approx 0.5757 \times 0.2116 = 0.1218$

Updated after $t = 4$: $\quad h_4^{(2)} \approx 0.1218, \quad C_4^{(2)} \approx 0.2148$

Time Step $t = 5$

Inputs: $h_4^{(2)} \approx 0.1218, \quad C_4^{(2)} \approx 0.2148, \quad$ input from Unit 1: $h_5^{(1)} \approx 0.2246$

Forget gate

- Linear combination: $0.33 \times 0.1218 + (-0.21) \times 0.2246 + 0.07 = 0.0402 + (-0.0472) + 0.07 = 0.0630$
- $f_5^{(2)} = \sigma(0.0630) \approx 0.5158$

Input gate (sigmoid)

- Linear combination: $0.29 \times 0.1218 + 0.38 \times 0.2246 + 0.12 = 0.0353 + 0.0853 + 0.12 = 0.2407$
- $i_5^{(2)} = \sigma(0.2407) \approx 0.5599$

Candidate

- Linear combination: $0.18 \times 0.1218 + 0.27 \times 0.2246 + 0.14 = 0.0219 + 0.0606 + 0.14 = 0.2226$
- $\tilde{C}_5^{(2)} = \tanh(0.2226) \approx 0.2190$

Output gate

- Linear combination: $0.41 \times 0.1218 + 0.24 \times 0.2246 + 0.21 = 0.0499 + 0.0539 + 0.21 = 0.3138$
- $o_5^{(2)} = \sigma(0.3138) \approx 0.5778$

Cell state update

- $C_5^{(2)} = (0.5158 \times 0.2148) + (0.5599 \times 0.2190) = 0.1108 + 0.1226 = 0.2334$

Hidden state update

- $h_5^{(2)} = 0.5778 \times \tanh(0.2334) \approx 0.5778 \times 0.2292 = 0.1325$

Updated after $t = 5$: $\quad h_5^{(2)} \approx 0.1325, \quad C_5^{(2)} \approx 0.2334$

Final updated values for Unit 2 after $t = 5$: $\quad h_5^{(2)} \approx 0.1325, \quad C_5^{(2)} \approx 0.2334$

Network Output and Anomaly Score

The output of the anomaly detection network is the final hidden state of the top unit, $h_5^{(2)} \approx 0.1325$. If the expected "normal" value for this sequence is y, the anomaly score is the squared prediction error

$$\text{score} = \tfrac{1}{2}\left(h_5^{(2)} - y\right)^2,$$

and the sequence is flagged as anomalous when the score exceeds a chosen threshold. This is exactly the loss function used in the backpropagation document, so training the network on normal sequences drives the score toward zero for normal data.

This completes the full forward pass of the two-unit stacked network. Every intermediate linear combination, activation, and state update has been shown with explicit arithmetic for all 5 time steps of both units. At each time step, Unit 1 is evaluated first (Part I), its hidden state $h_t^{(1)}$ is fed into Unit 2, and both units carry their own states forward to the next time step.